VIRAJ VAGHASIA

Los Angeles, CA | +1 213-803-5009 | virajvaghasiamv907@gmail.com | LinkedIn | GitHub | Portfolio

M.S. Computer Science candidate (May 2027) with three years of production software engineering experience, building backend services, data pipelines, and document retrieval systems.

Education

University of Southern California, Los Angeles, CA — Aug 2025 – May 2027
Master of Science in Computer Science — GPA: 3.86/4.00

Dwarkadas J. Sanghvi College of Engineering, Mumbai, India — Aug 2018 – May 2022
Bachelor of Engineering, Information Technology — GPA: 9.38/10

Coursework: Analysis of Algorithms, Information Retrieval, Database Systems, Applied Natural Language Processing

Technical Skills

Languages: Python, JavaScript, TypeScript, SQL, Bash, C++

Backend & Data: FastAPI, Django, Flask, Node.js, Express, REST APIs, PostgreSQL, MySQL, MongoDB, Redis, SQLite, ETL and batch pipelines

Frontend: React, Next.js, Vue.js, Redux, Tailwind CSS

Infrastructure & Testing: Docker, AWS, GitHub Actions (CI/CD), Git, pytest, Playwright, Modal

Search & Retrieval: Dense embeddings, BM25, hybrid ranking, cross-encoder reranking, evaluation harnesses, Qdrant, vLLM, Ollama

Experience

Research Assistant, USC Information Sciences Institute — May 2026 – Present

Geological literature acquisition and classification pipeline for a critical-minerals research project, Los Angeles, CA

- Built a **restartable document-acquisition service** in Python checkpointed on a SQLite ledger rather than a run flag, pairing an open-access tier with an authenticated Playwright tier through the USC library proxy; the ledger records **33,169** downloaded articles across **10** journals.
- Replaced per-publisher scrapers with **declarative JSON recipes** covering 4 platforms, plus a generator that proposes extraction rules from one open-access article, where deterministic checks decide what ships rather than the model.
- Classified crawl failures into **login redirects, bot challenges and rate-limit blocks** instead of retrying blindly, with escalating **3-to-30-minute backoff** and a halt after three consecutive blocks, because an institutional IP block is terminal rather than retryable.
- Diagnosed **silent data-loss defects**, including discarded supplementary files and an ignored year filter, by comparing files on content hash after filename comparison produced false mismatches; grew the pytest suite to **484** tests.

Senior Software Engineer, Decimal Point Analytics, Mumbai, India — Jul 2022 – Jul 2025

- Developed a **real-time financial index platform** in React and FastAPI tracking multiple indices simultaneously, replacing static analyst reports with interactive time-series views.
- Implemented a **maker-checker review workflow** in Django separating the analyst entering data from the reviewer approving it, blocking unreviewed records from publication.
- Led a **Loan Management System** and an **ESG questionnaire portal** across several engineers, onboarding new joiners and gathering requirements from analysts; designed the questionnaire schema so business users could change questions without a code deploy (Vue.js, Next.js, FastAPI).

Projects

SQLAlchemy Upgrade Agent | GitHub | Python, FastAPI, Qdrant, GitHub Actions, Modal — Jun 2026 – Sep 2026

- Shipped a **deployed question-answering service** over SQLAlchemy 1.4 → 2.0 migration documentation that cites a source page for every claim and declines when the retrieved pages do not answer.
- Built **hybrid retrieval** over **3,284** passages using dense embeddings with BM25, duplicate collapsing and cross-encoder reranking, raising correct-page-in-top-5 from **0.49 to 0.64** on a **100-question** hand-verified set, measured as paired wins rather than an average.
- Verified answers by **executing each claim against SQLAlchemy 2.0.51** instead of trusting a model's self-report (**47 of 51** correct), and gated retrieval quality in **GitHub Actions**, which blocked a pull request removing the reranker.
- **Red-teamed the deployed service** against prompt injection over **90** pre-registered attempts, cutting obeyed attempts **16 → 10** with fencing (6 fixed, 0 broken, $p = 0.031$), and shipped none of it because the bar written before the run was ≤ 5.
- Rejected two proposed retrieval improvements on their own measurements: one dropped recall from 0.64 to 0.58 and was reverted, the other was cancelled before implementation once the defect it assumed was shown not to exist.